

Computer Vision with Deep Learning

Lecture 16: Projective geometry and calibration

slides credit: Ioannis Gkioulekas, Lana Lazebnik, Raymond Yeh, David Lindell, (and whoever mentioned in specific slides)

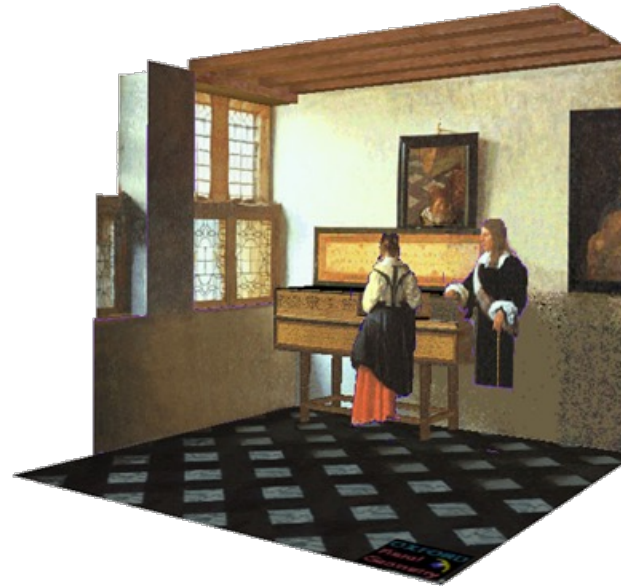
Announcements

- Most project proposals fine
 - will reach out with suggestions
- HW3 due April 10
- Midterm grading in progress

Our goal: recover 3D information from images

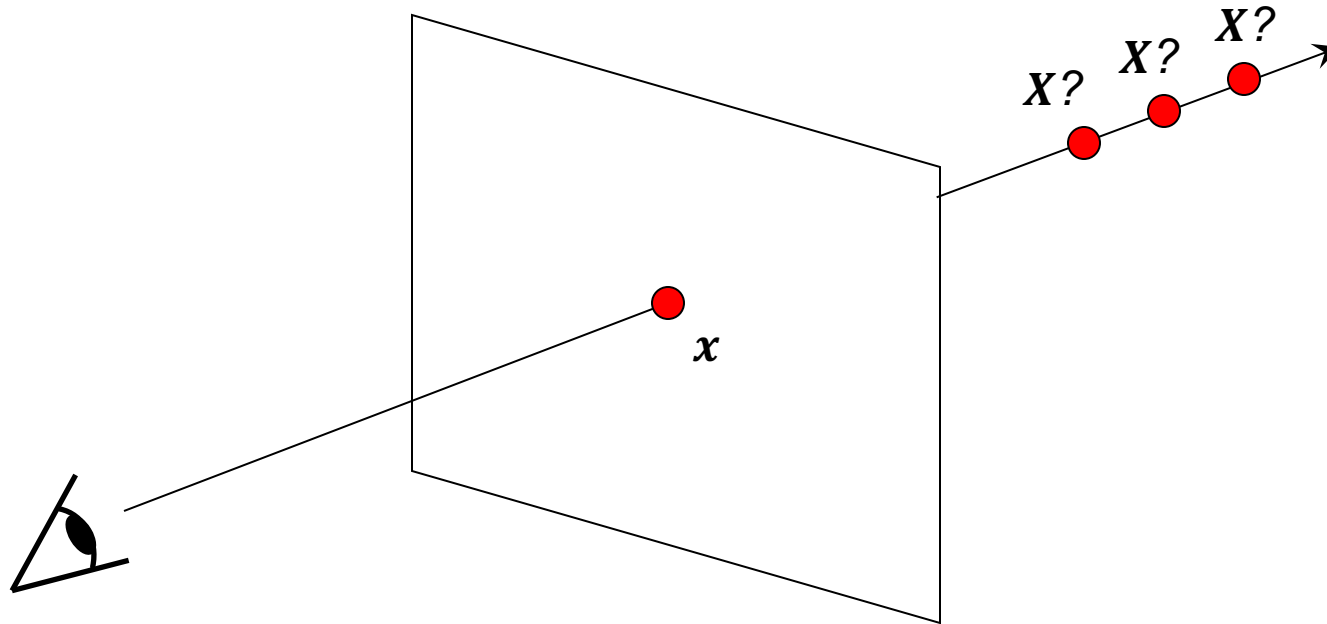


J. Vermeer, Music Lesson, 1662



A. Criminisi et al. [Bringing Pictorial Space to Life: computer techniques for the analysis of paintings](#),
Proc. Computers and the History of Art, 2002

Single-view ambiguity

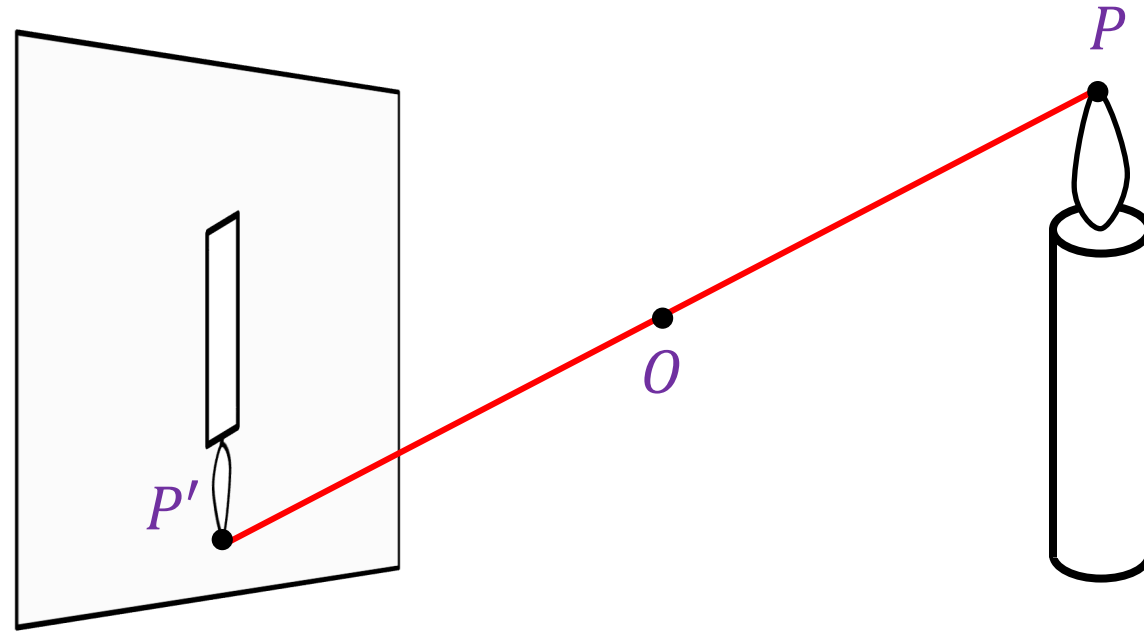


Overview

- pinhole projection model
- properties of projection
- perspective projection
- orthographic projection
- camera calibration

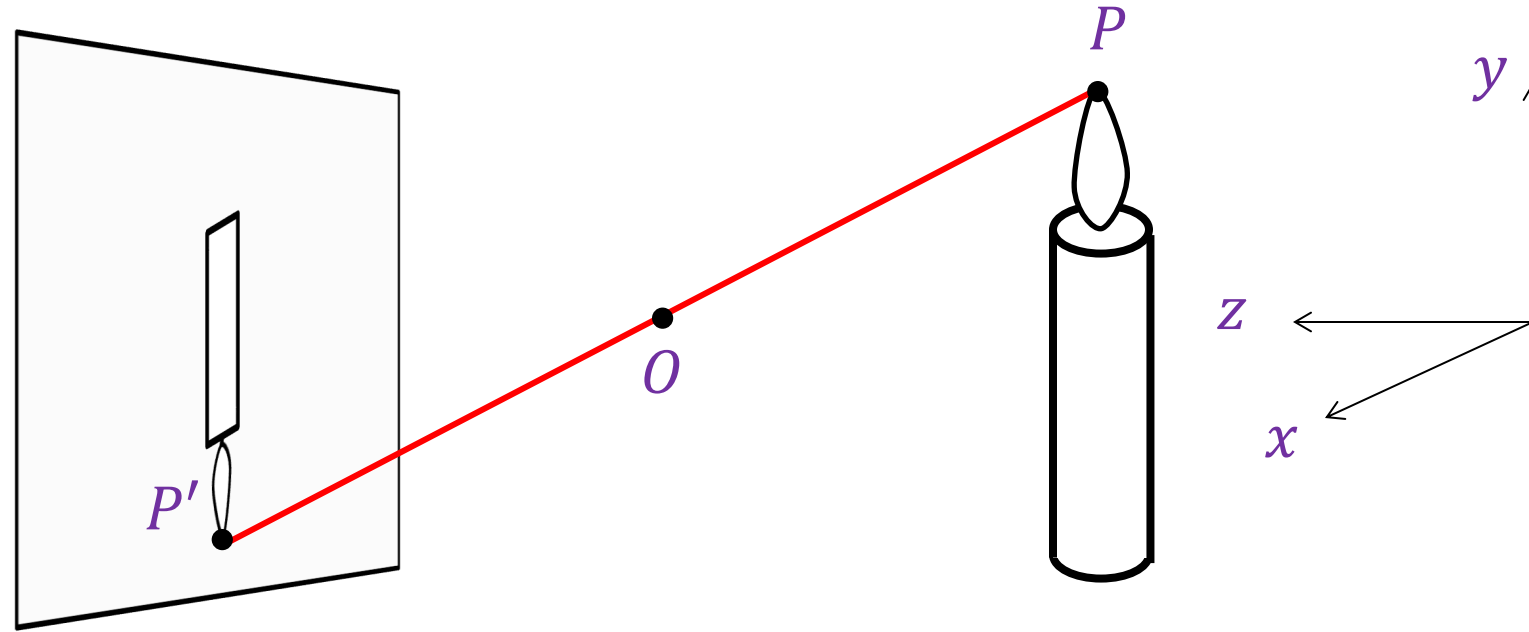
pinhole projection model

Modeling projection



- How do we find the projection P' of a scene point P ?
 - Form the **visual ray** connecting P to the camera center O and find where it intersects the image plane
- All scene points that lie on this visual ray have the same projection in the image
- Are there scene points for which this projection is undefined?

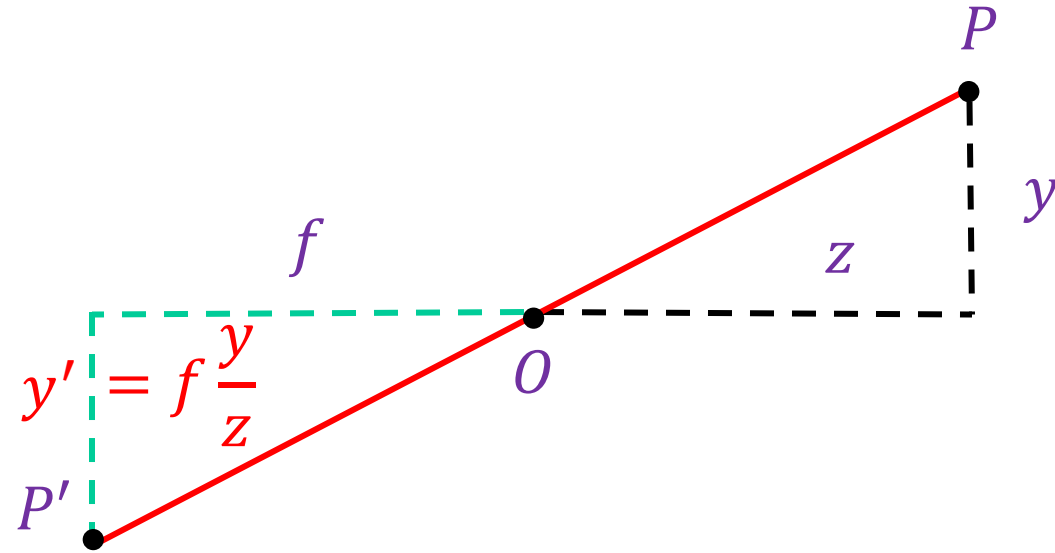
Modeling projection



Canonical coordinate system

- The optical center (O) is at the origin
- The z axis is the *optical axis* perpendicular to the image plane
- The xy plane is parallel to the image plane, x and y axes are horizontal and vertical directions of the image plane

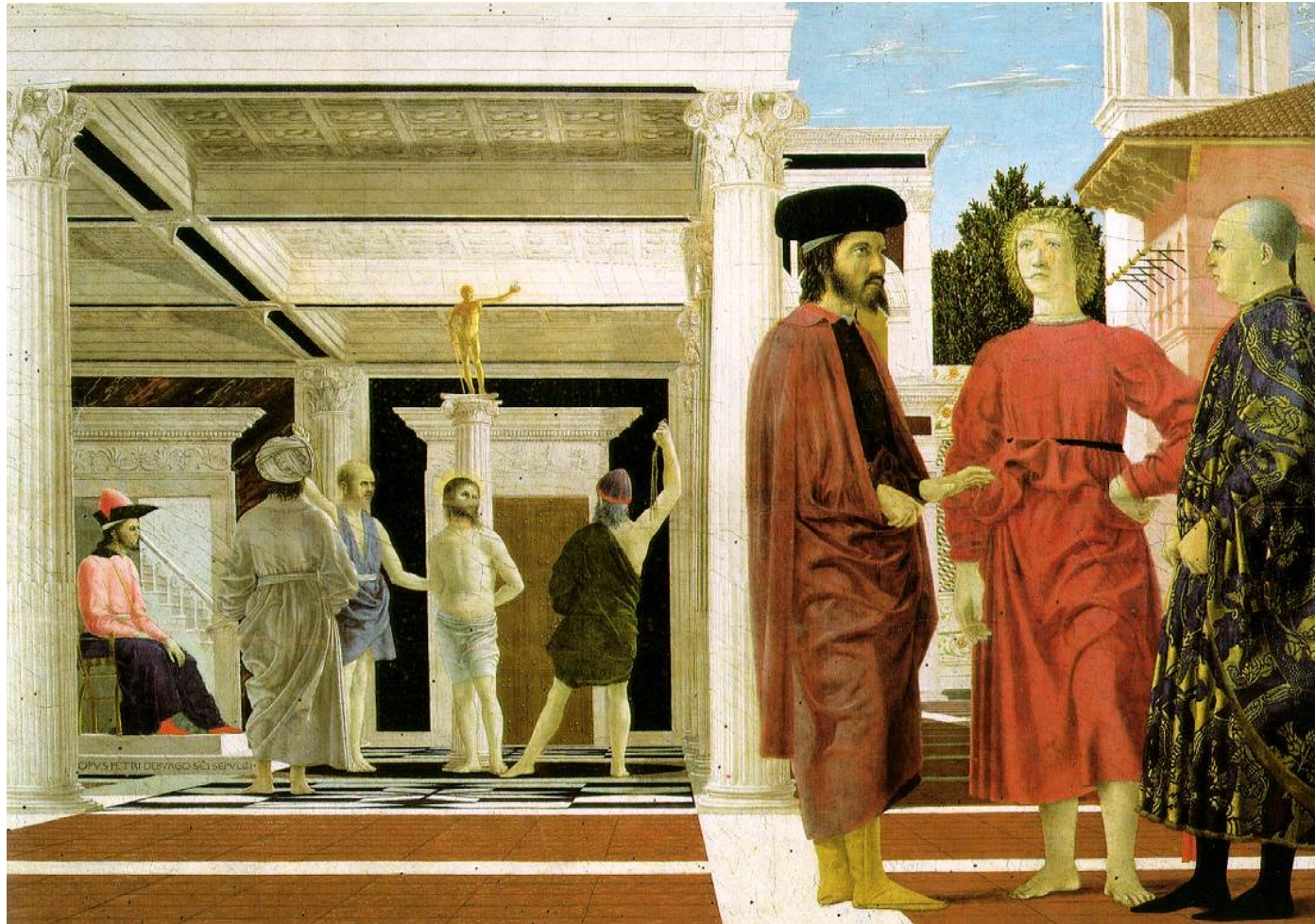
Deriving projected coordinates



$$(x, y, z) \rightarrow \left(f \frac{x}{z}, f \frac{y}{z} \right)$$

properties of projection

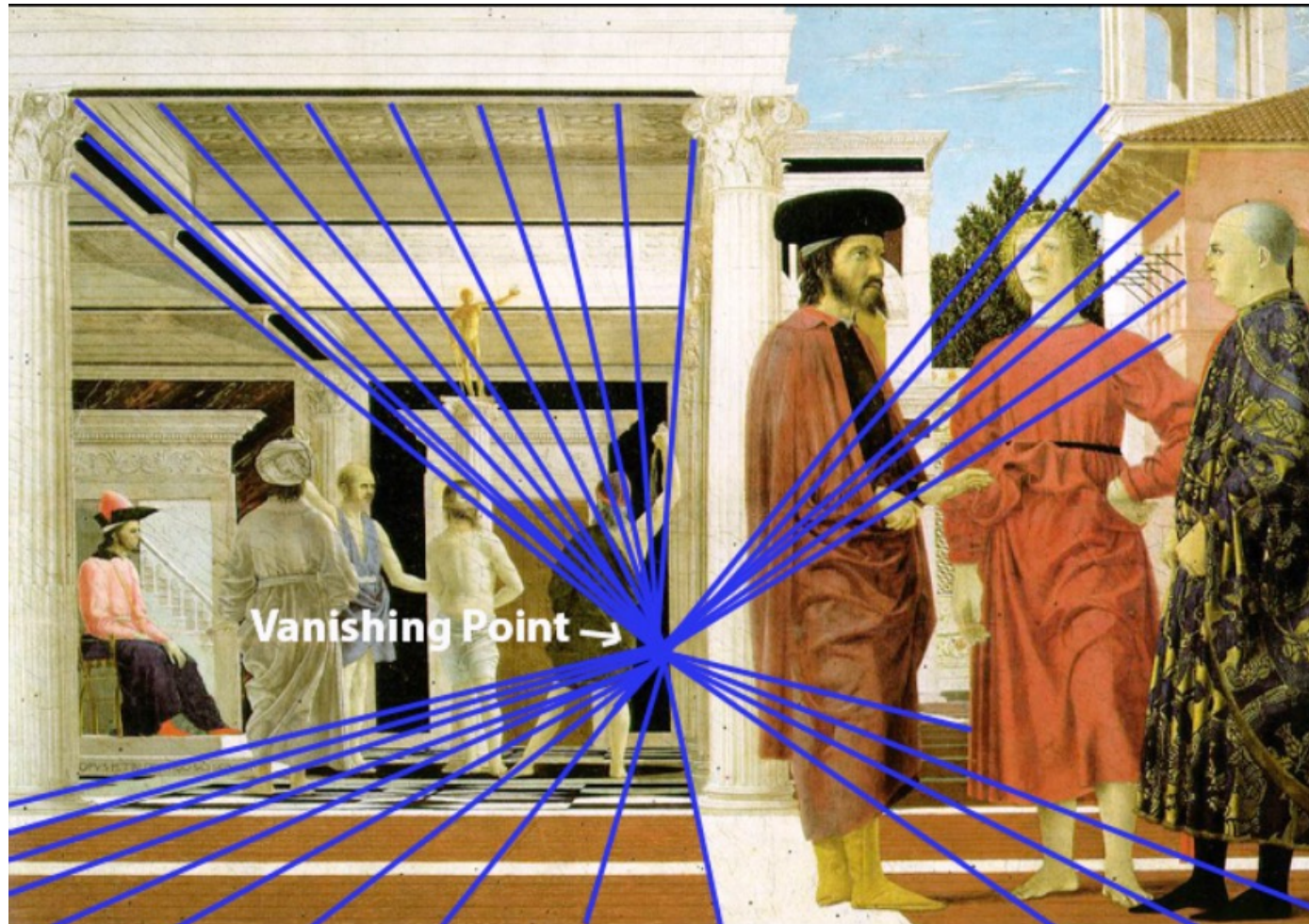
Projection of lines



Piero della Francesca, *Flagellation of Christ*, 1455-1460

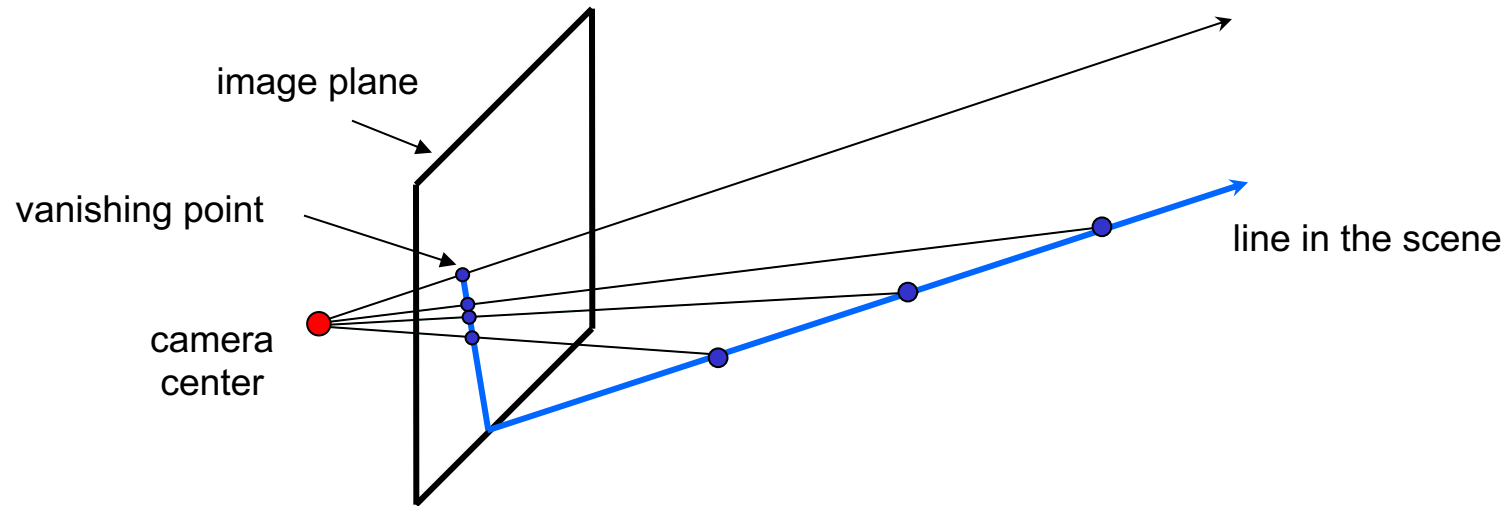
Vanishing points

- Parallel lines meet at a *vanishing point*

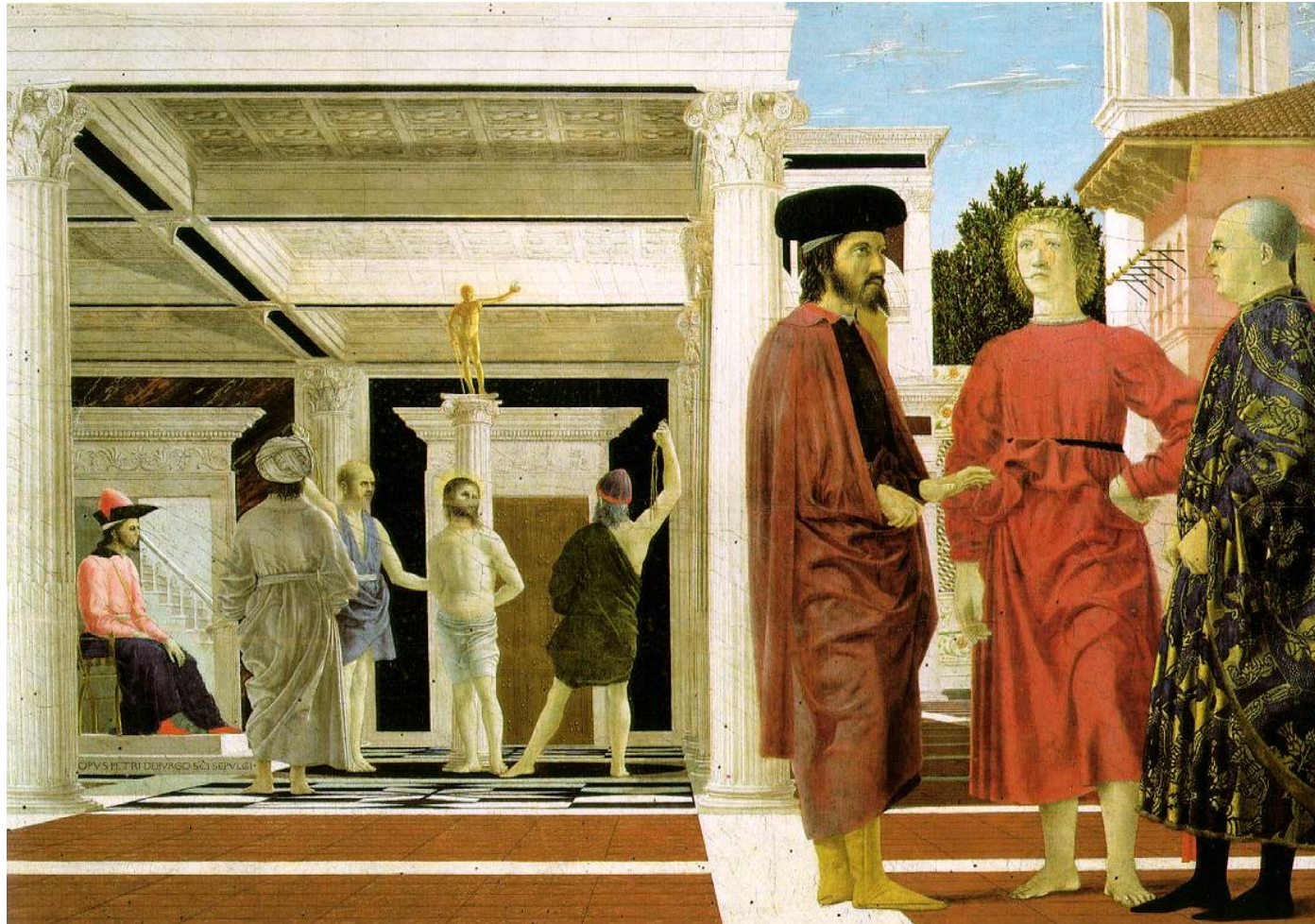


Piero della Francesca, *Flagellation of Christ*, 1455-1460

Constructing the vanishing point of a line



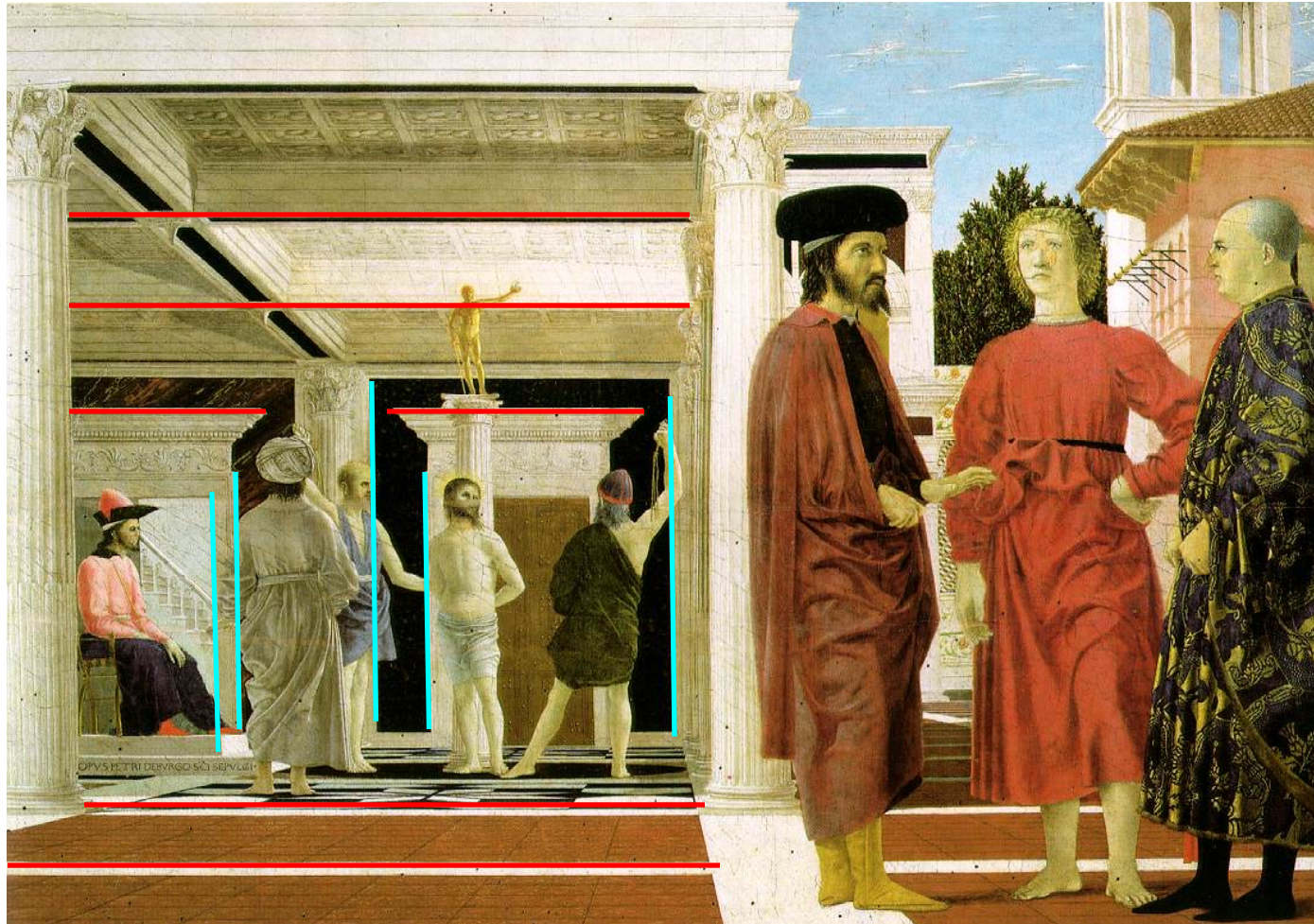
Do *all* parallel lines converge to a vanishing point?



Piero della Francesca, *Flagellation of Christ*, 1455-1460

Do *all* parallel lines converge to a vanishing point?

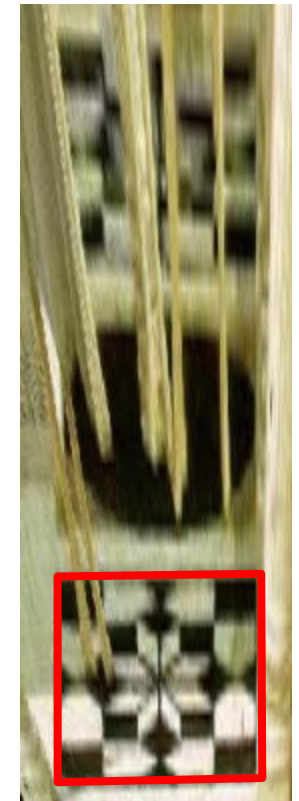
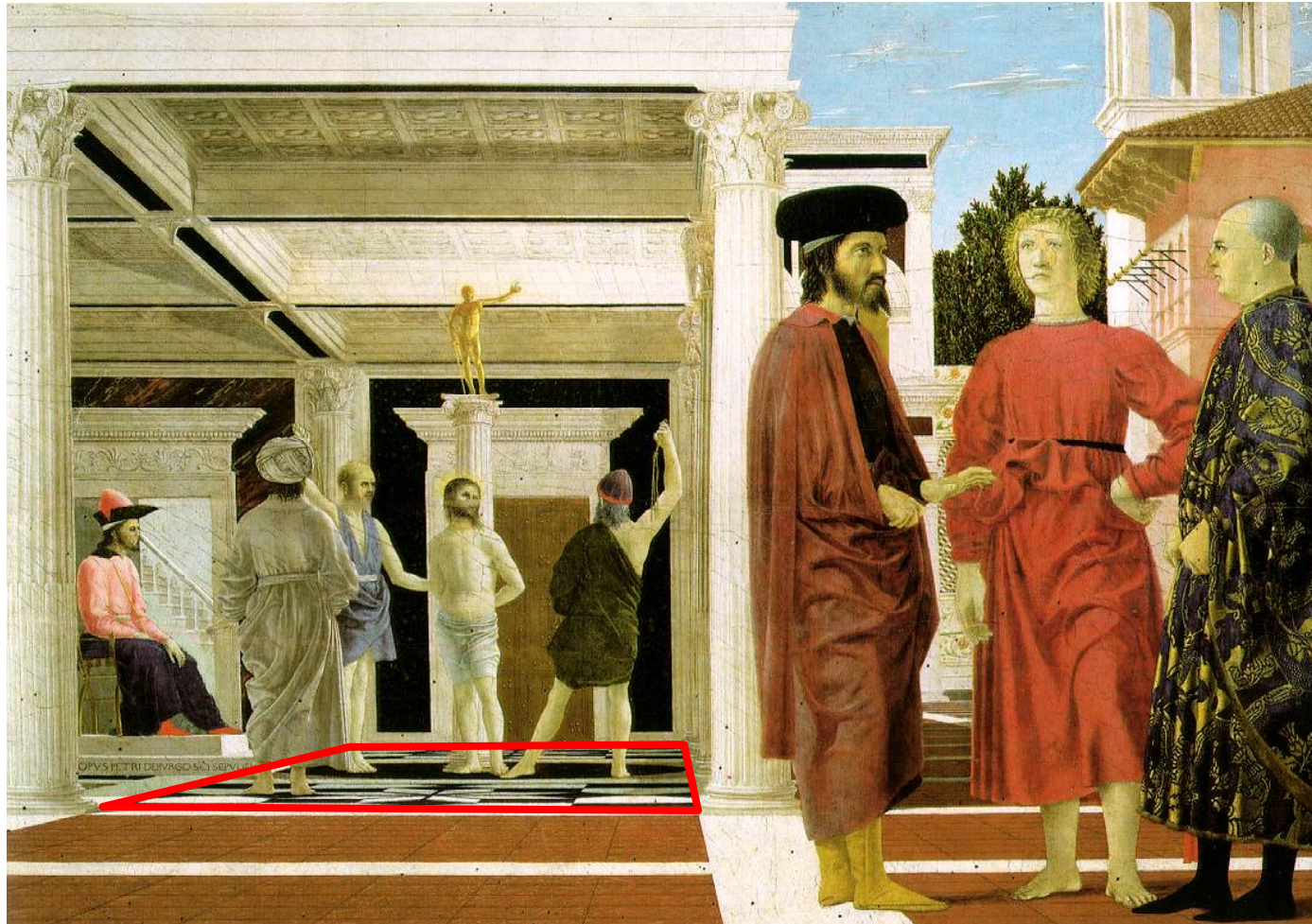
- Not parallel lines that are also parallel to the image plane!



Piero della Francesca, *Flagellation of Christ*, 1455-1460

Projection of planes

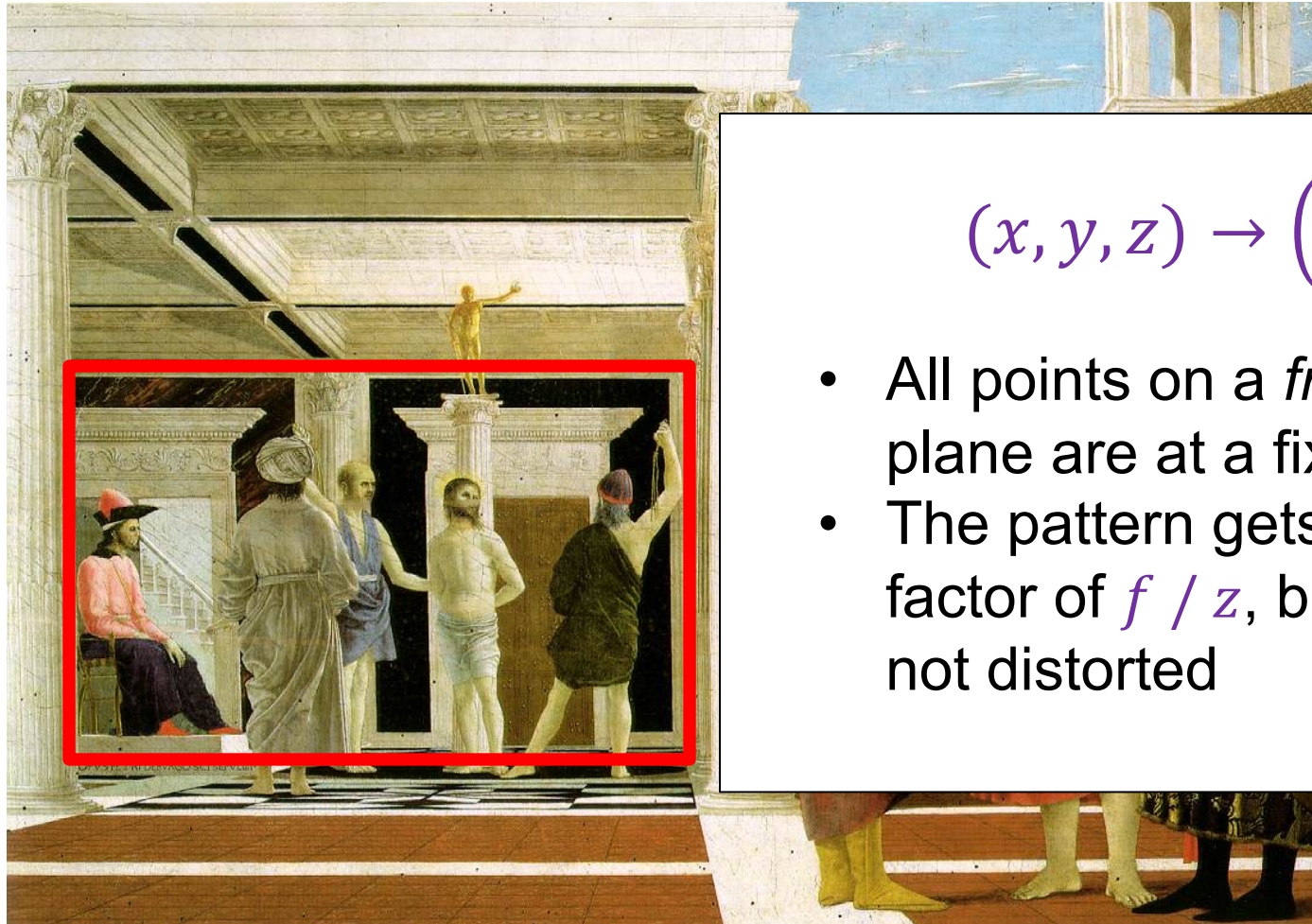
- Patterns on *non-fronto-parallel* planes are distorted by a 2D homography



Piero della Francesca, *Flagellation of Christ*, 1455-1460

Projection of planes

- What about fronto-parallel planes?



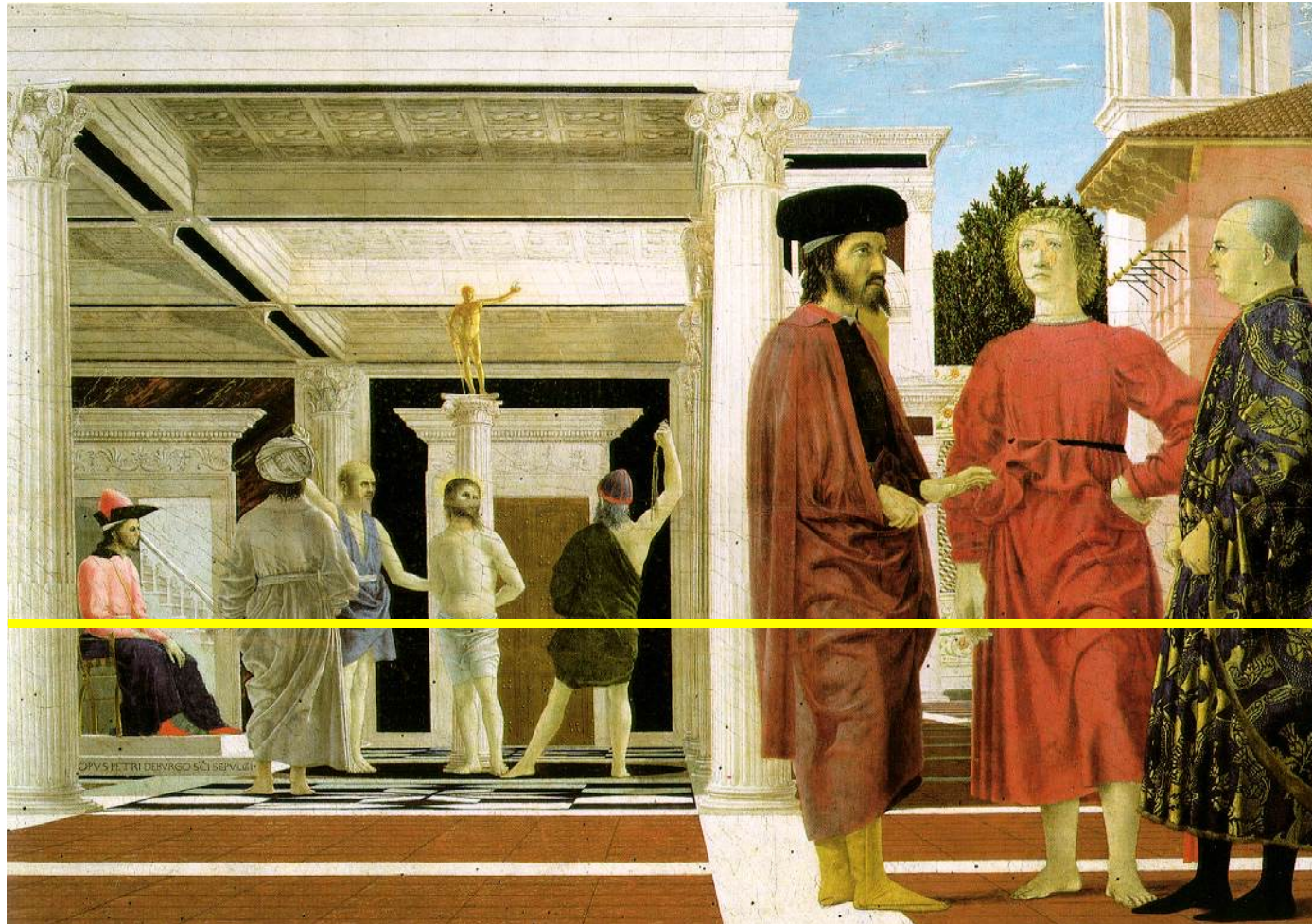
$$(x, y, z) \rightarrow \left(f \frac{x}{z}, f \frac{y}{z}\right)$$

- All points on a *fronto-parallel* plane are at a fixed depth z
- The pattern gets scaled by a factor of f / z , but is otherwise not distorted

Piero della Francesca, *Flagellation of Christ*, 1455-1460

Projection of planes

- Non fronto-parallel planes have vanishing lines

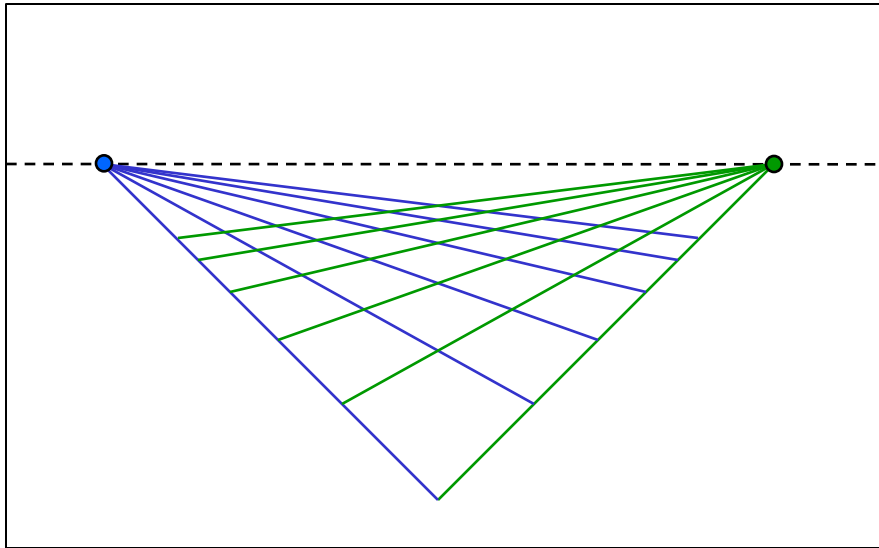


Vanishing line of
the ground plane

Piero della Francesca, *Flagellation of Christ*, 1455-1460

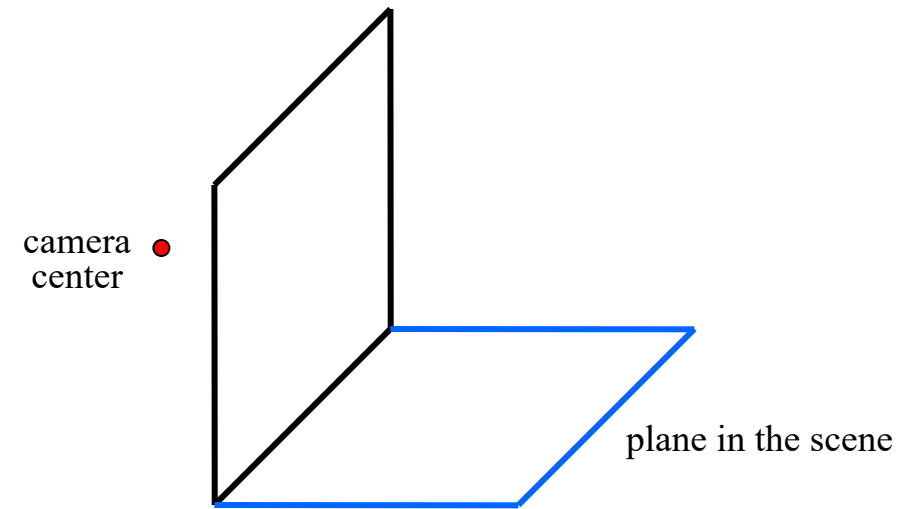
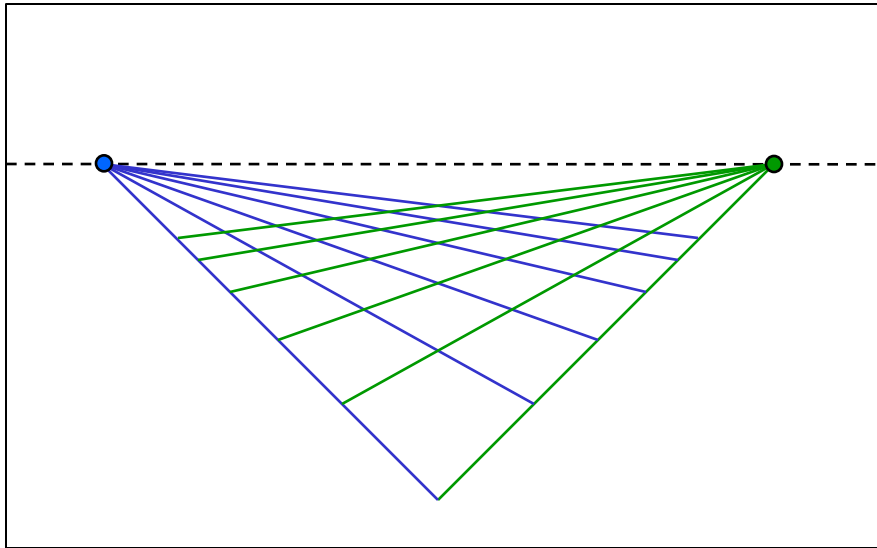
Constructing the vanishing line of a plane

- Each family of parallel planes is associated with a vanishing line in the image



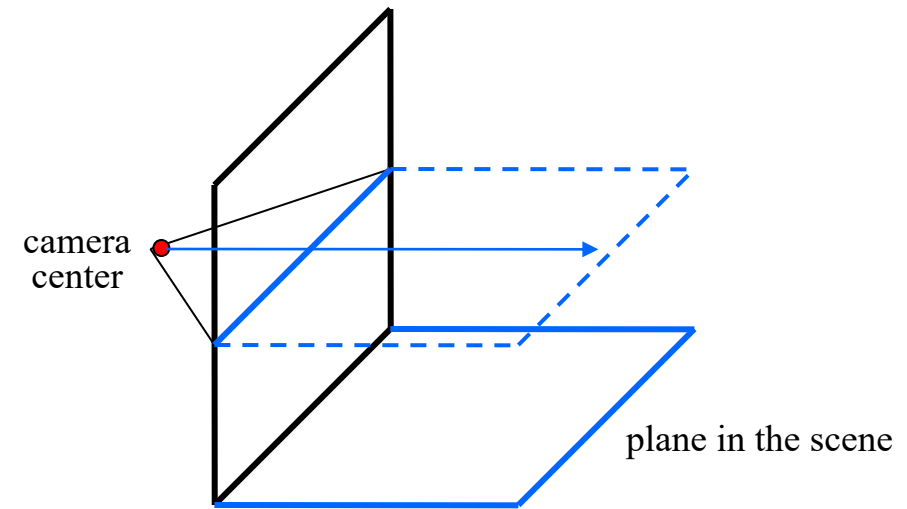
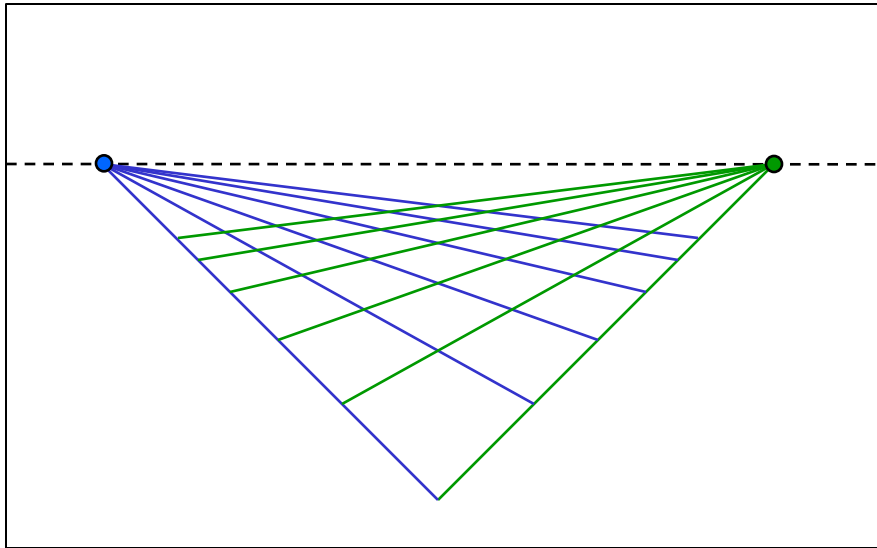
Constructing the vanishing line of a plane

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Constructing the vanishing line of a plane

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HORIZON



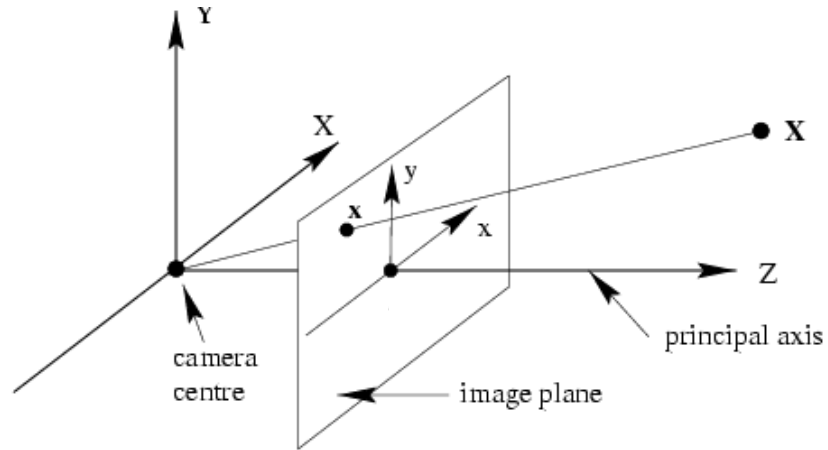
Generative models and vanishing lines



[source](#)

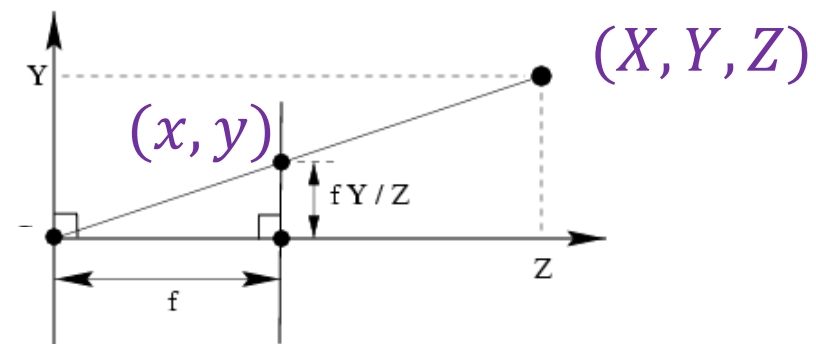
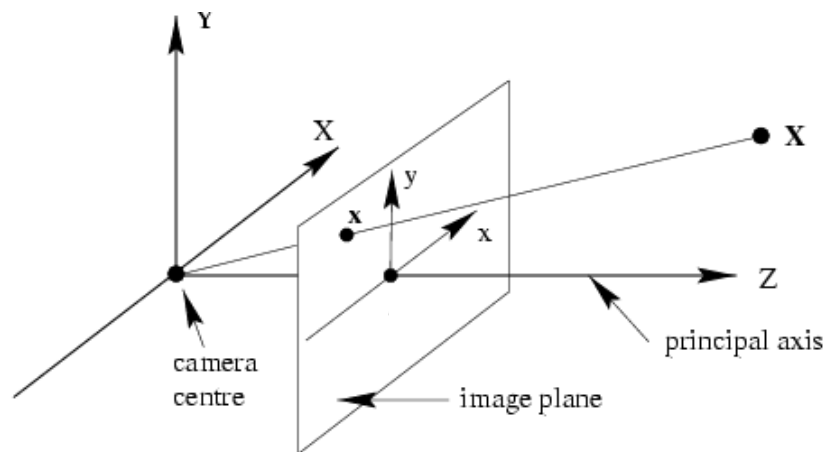
perspective and orthographic projection

Review: Perspective projection



- **Normalized (camera) coordinate system:** camera center is at the origin, the *principal axis* is the z -axis, x and y axes of the image plane are parallel to x and y axes of the world

Review: Perspective projection



$$x = f \frac{X}{Z}, y = f \frac{Y}{Z}$$

Review: Homogeneous coordinates

- To form homogeneous coordinates from Euclidean coordinates, append 1 as the last entry:

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

homogeneous *image*
coordinates

$$(X, Y, Z) \Rightarrow \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

homogeneous *scene*
coordinates

- To convert *from* homogeneous coordinates, divide by the last entry:

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

$$\begin{bmatrix} X \\ Y \\ Z \\ W \end{bmatrix} \Rightarrow (X/W, Y/W, Z/W)$$

In homogeneous coordinates, all scalar multiples represent the same point!

Perspective projection matrix

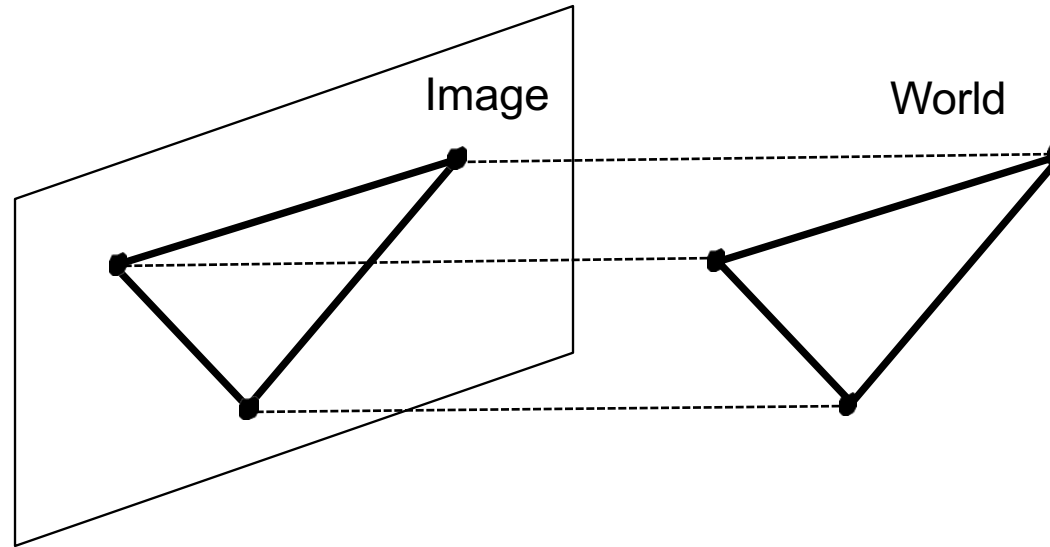
- Projection is a matrix multiplication using homogeneous coordinates:

$$\begin{bmatrix} \\ \\ \\ \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} \\ \\ \\ \end{bmatrix} \Rightarrow \left(f \frac{X}{Z}, f \frac{Y}{Z} \right)$$

divide by the third coordinate

Orthographic projection

- Special case of perspective projection
 - Distance from center of projection to image plane is **infinite**
 - Also called “parallel projection”

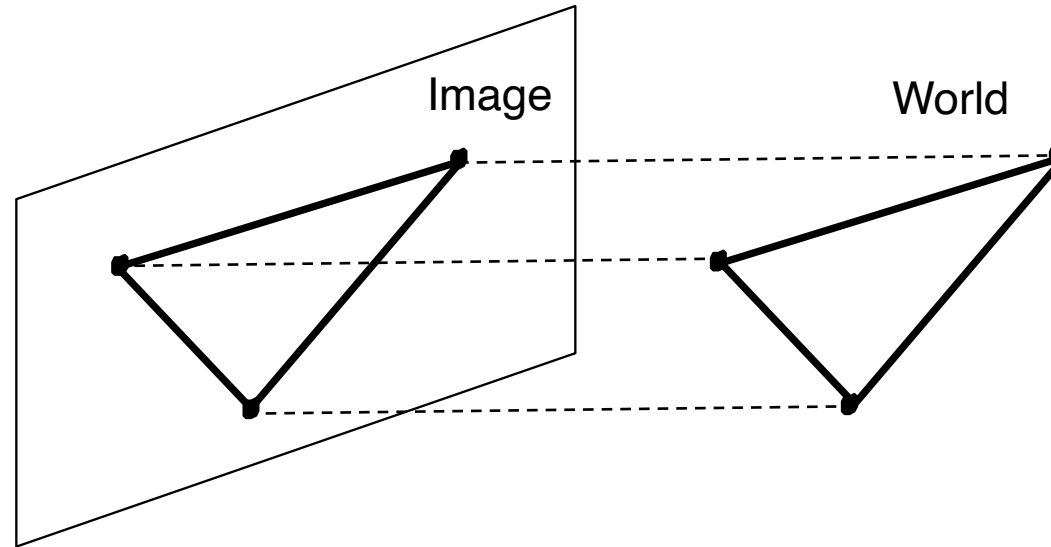


Orthographic projection

- Special case of perspective projection
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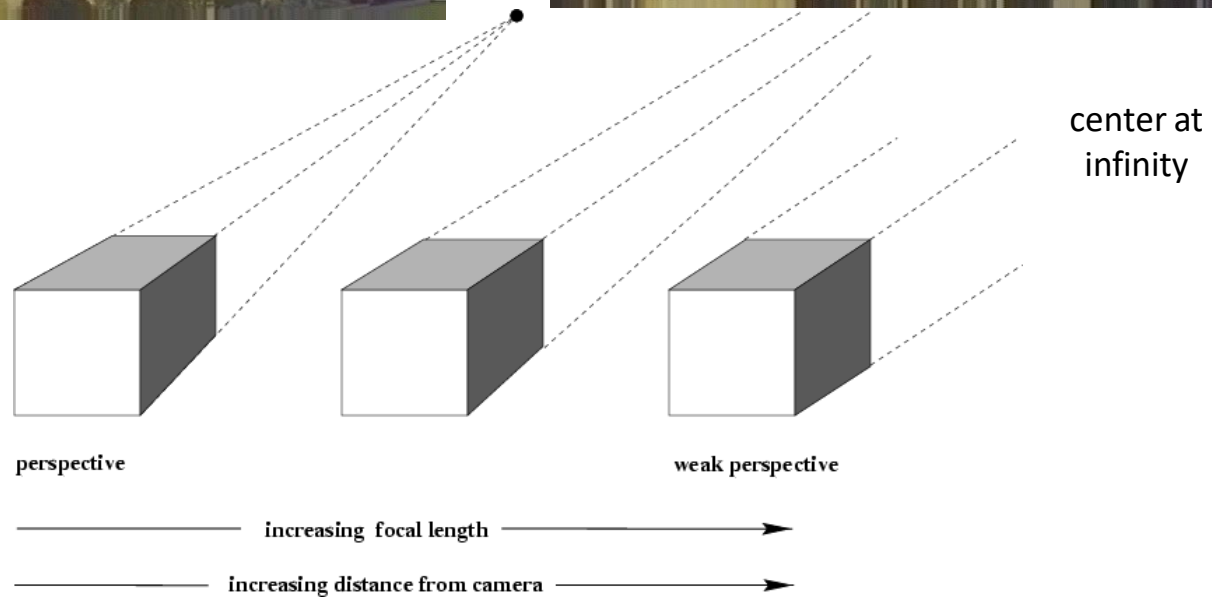
Orthographic projection matrix



- Assuming projection along the z axis, what's the matrix?

$$\begin{bmatrix} \\ \\ \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} \\ \end{bmatrix}$$

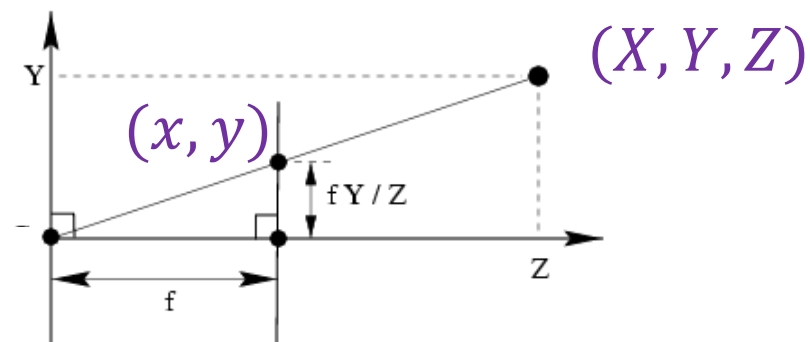
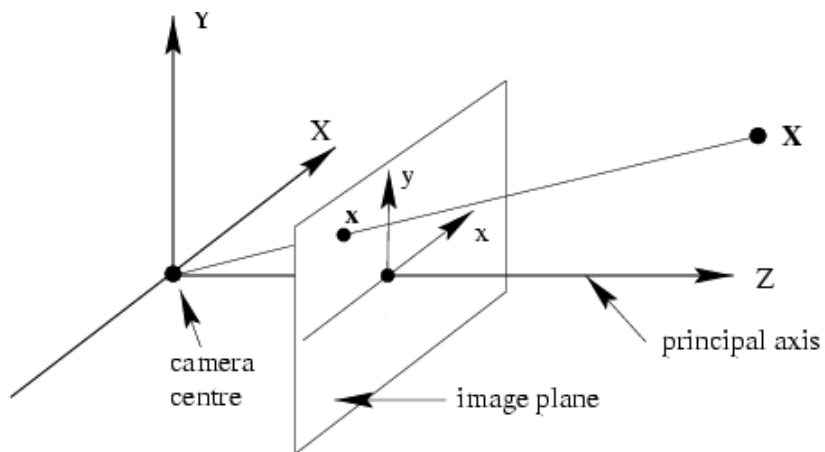
Approximating an orthographic camera



Source: Hartley & Zisserman

camera calibration

Perspective projection in normalized coordinates



$$x = f \frac{X}{Z}, y = f \frac{Y}{Z}$$

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \cong \begin{pmatrix} fX \\ fY \\ Z \end{pmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

$\mathbf{x}$

Homogeneous
coord. vec. of
image point

$\mathbf{P}$

Camera
projection
matrix

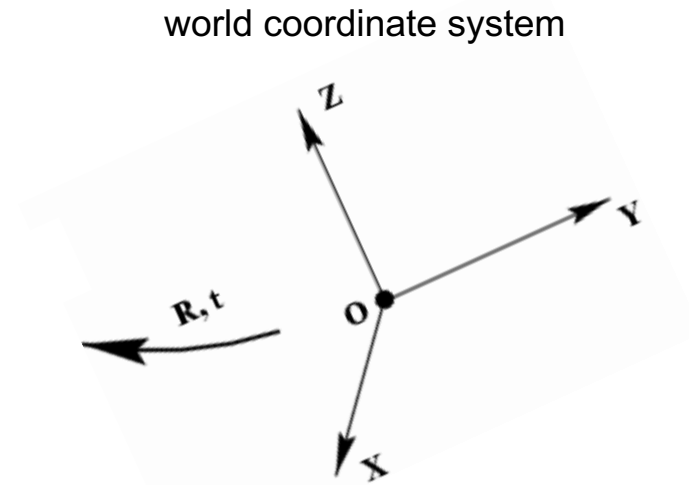
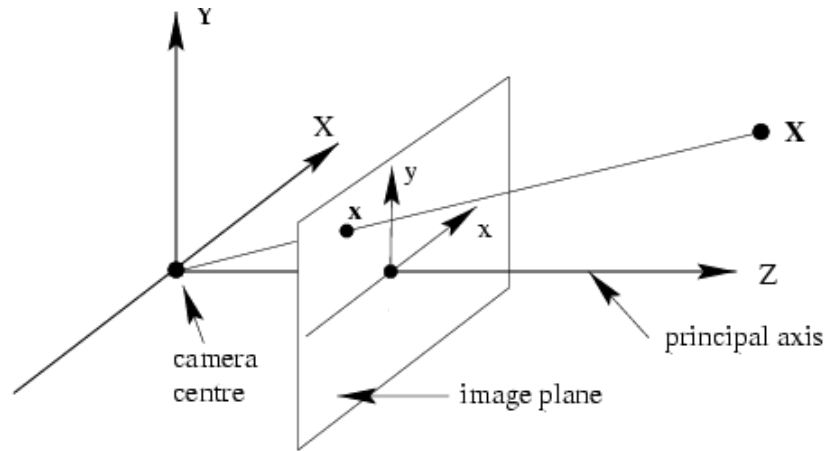
$\mathbf{X}$

Homogeneous
coord. vec. of 3D
point

$$\mathbf{x} \cong \mathbf{P}\mathbf{X}$$

Equality up to scale

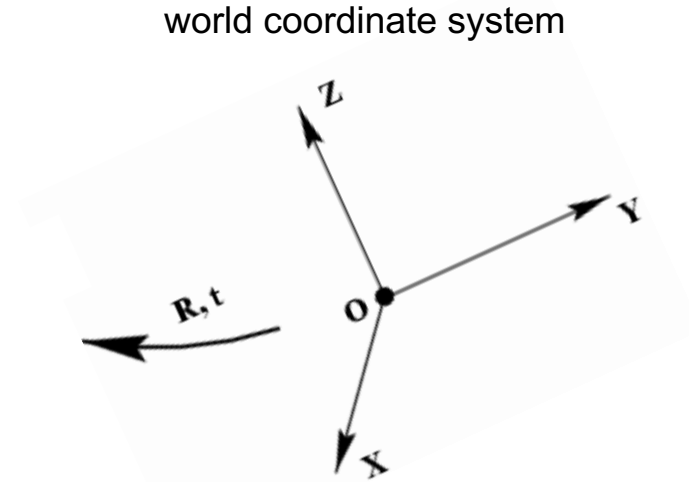
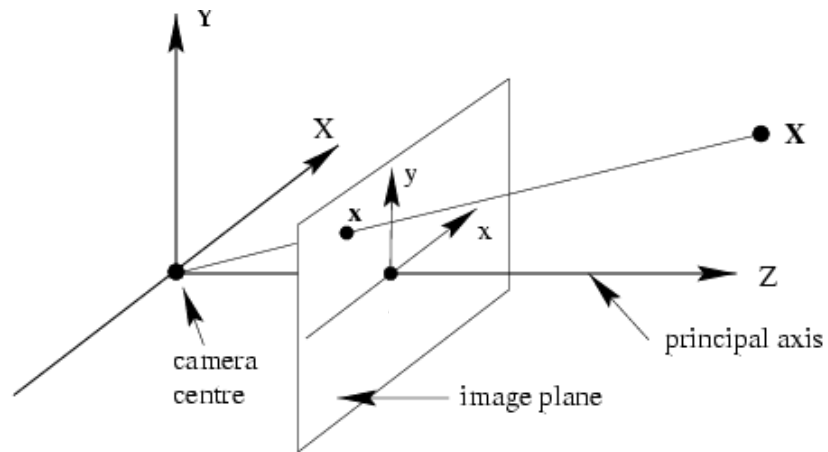
Camera parameters and calibration



- **Camera calibration:** figuring out transformation from *world* coordinate system to *image* coordinate system

$$\begin{array}{c}
 \begin{pmatrix} \text{2D point} \\ (3 \times 1) \end{pmatrix} \\
 \mathbf{x}
 \end{array}
 \cong
 \begin{array}{c}
 \begin{pmatrix} \text{Camera to pixel coord. trans. matrix} \\ \mathbf{K} \ (3 \times 3) \end{pmatrix} \\
 \text{Intrinsic camera parameters: principal point, scaling factors}
 \end{array}
 \begin{array}{c}
 \begin{pmatrix} \text{Canonical projection matrix} \\ \mathbf{[I \mid 0]} \ (3 \times 4) \end{pmatrix} \\
 \text{Extrinsic camera parameters: rotation, translation}
 \end{array}
 \begin{array}{c}
 \begin{pmatrix} \text{World to camera coord. trans. matrix} \\ \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \ (4 \times 4) \end{pmatrix} \\
 \mathbf{X}
 \end{array}
 \begin{array}{c}
 \begin{pmatrix} \text{3D point} \\ (4 \times 1) \end{pmatrix}
 \end{array}$$

Camera parameters and calibration

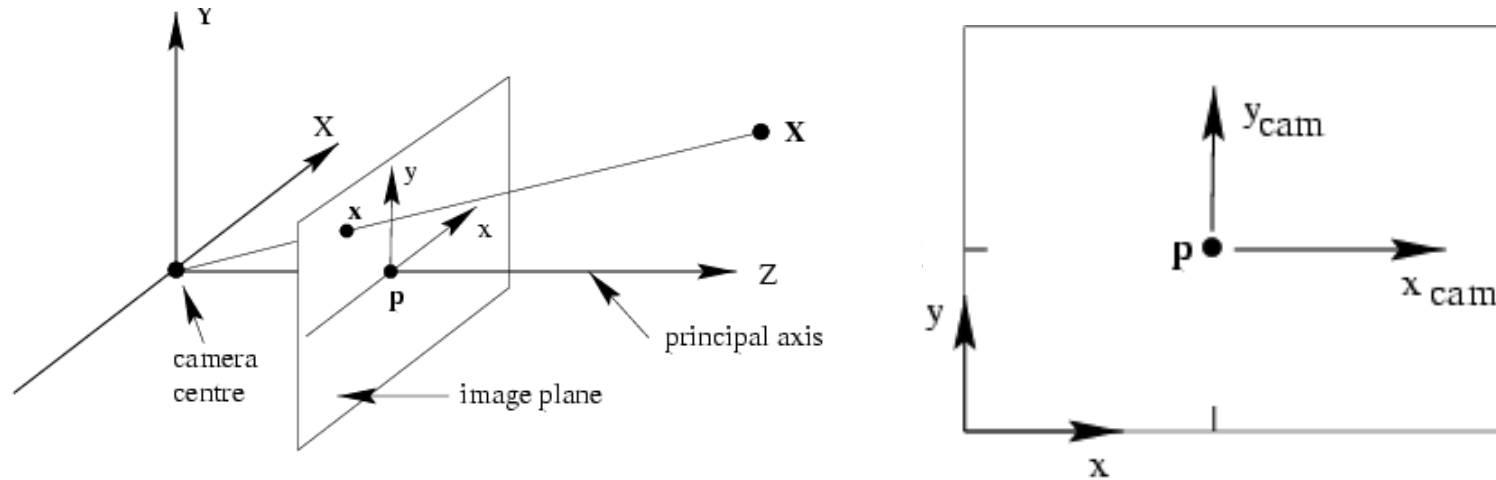


- **Camera calibration:** figuring out transformation from *world* coordinate system to *image* coordinate system

$$\begin{array}{c}
 \begin{pmatrix} \text{2D point} \\ \text{(3x1)} \end{pmatrix} \\
 \mathbf{x}
 \end{array}
 \cong
 \underbrace{
 \begin{pmatrix} \text{Camera to pixel coord. trans. matrix} \\ \mathbf{K} \text{ (3x3)} \end{pmatrix}
 \begin{pmatrix} \text{Canonical projection matrix} \\ \mathbf{[I | 0]} \text{ (3x4)} \end{pmatrix}
 \begin{pmatrix} \text{World to camera coord. trans. matrix} \\ \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \text{ (4x4)} \end{pmatrix}
 }_{\mathbf{P} = \mathbf{K}[\mathbf{R}|\mathbf{t}]}
 \begin{pmatrix} \text{3D point} \\ \text{(4x1)} \end{pmatrix}
 \mathbf{X}$$

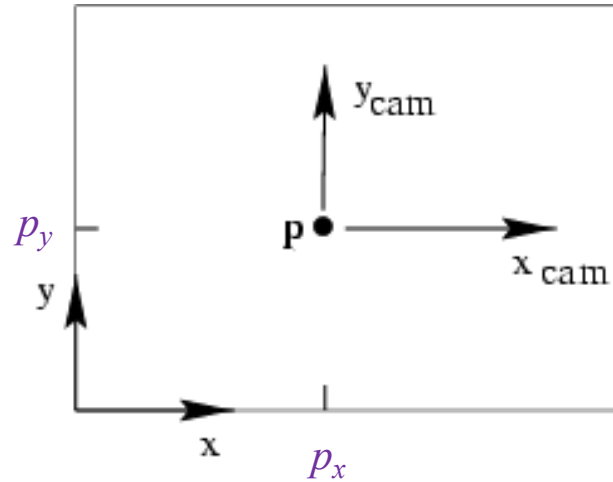
General camera projection matrix

Intrinsic parameters: Principal point



- Principal point (p): point where principal axis intersects the image plane
- In the *normalized* coordinate system, the **origin** of the image is at the **principal point**
- In the *image* coordinate system: the **origin** is in the **corner**

Intrinsic parameters: Principal point

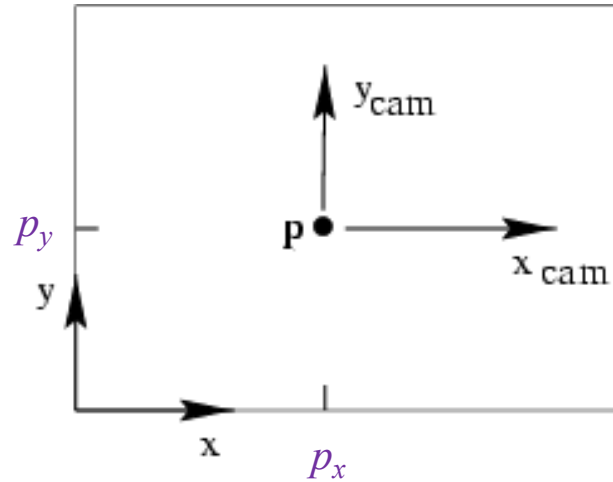


We want the principal point to map to (p_x, p_y) instead of $(0,0)$

$$x = f \frac{X}{Z} + p_x, \quad y = f \frac{Y}{Z} + p_y$$

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \cong \begin{pmatrix} fX + Zp_x \\ fY + Zp_y \\ Z \end{pmatrix} = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

Intrinsic parameters: Principal point



Principal point: (p_x, p_y)

$$\begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} f & 0 & p_x & 0 \\ 0 & f & p_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

calibration
matrix $\mathbf{K}$

Canonical
projection matrix
 $[\mathbf{I} \mid \mathbf{0}]$

$$\mathbf{P} = \mathbf{K}[\mathbf{I} \mid \mathbf{0}]$$

Intrinsic parameters: Principal point

- What are the units of the focal length f and principal point coordinates (p_x, p_y) ?
 - Same as world units – presumably metric units
- What units do we want for measuring image coordinates?
 - Pixel units
- Thus, we need to introduce *scaling factors* for mapping from world to pixel units

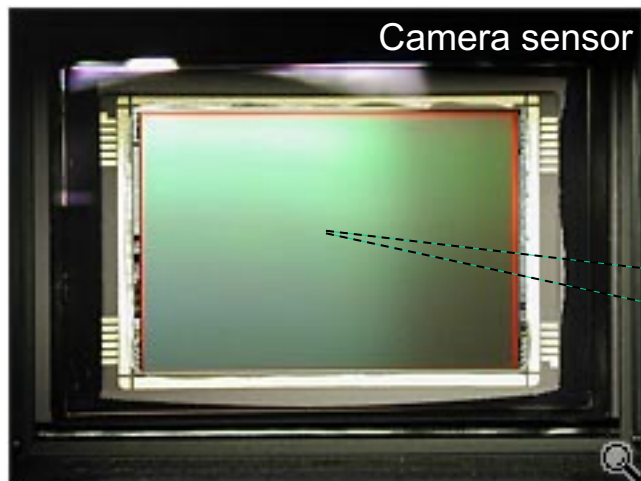
$$\begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} f & 0 & p_x & 0 \\ 0 & f & p_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

calibration
matrix $\mathbf{K}$

Canonical
projection matrix
 $[\mathbf{I} \mid \mathbf{0}]$

$$\mathbf{P} = \mathbf{K}[\mathbf{I} \mid \mathbf{0}]$$

Intrinsic parameters: Scaling factors



Camera sensor

m_x pixels/m in horizontal direction,
 m_y pixels/m in vertical direction

Pixel size (m): $\frac{1}{m_x} \times \frac{1}{m_y}$

Scaling factors

$$\begin{bmatrix} m_x & 0 & 0 \\ 0 & m_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

pixels/m

Calibration matrix
 K in metric units

$$\begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix}$$

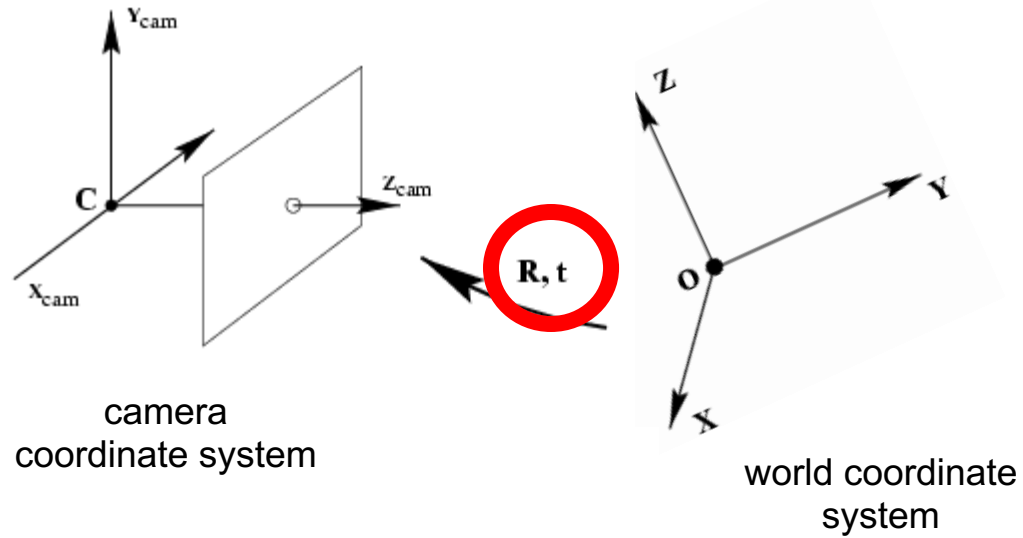
m

Calibration matrix
 K in pixel units

$$\begin{bmatrix} \alpha_x & 0 & \beta_x \\ 0 & \alpha_y & \beta_y \\ 0 & 0 & 1 \end{bmatrix}$$

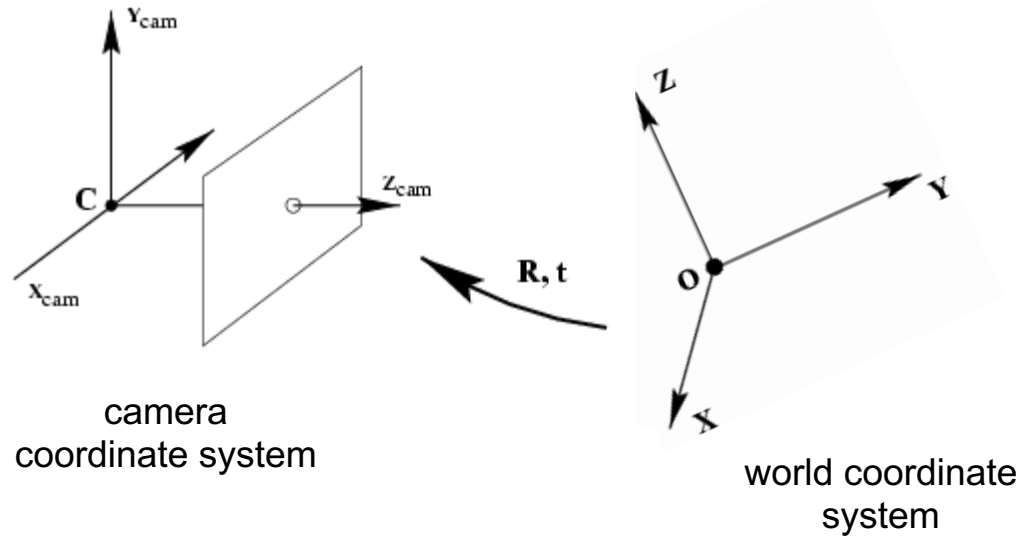
pixels

Extrinsic parameters: Rotation and translation



- In general, the camera coordinate frame will be related to the world coordinate frame by a rotation and a translation

Extrinsic parameters: Rotation and translation



- In general, the camera coordinate frame will be related to the world coordinate frame by a rotation and a translation

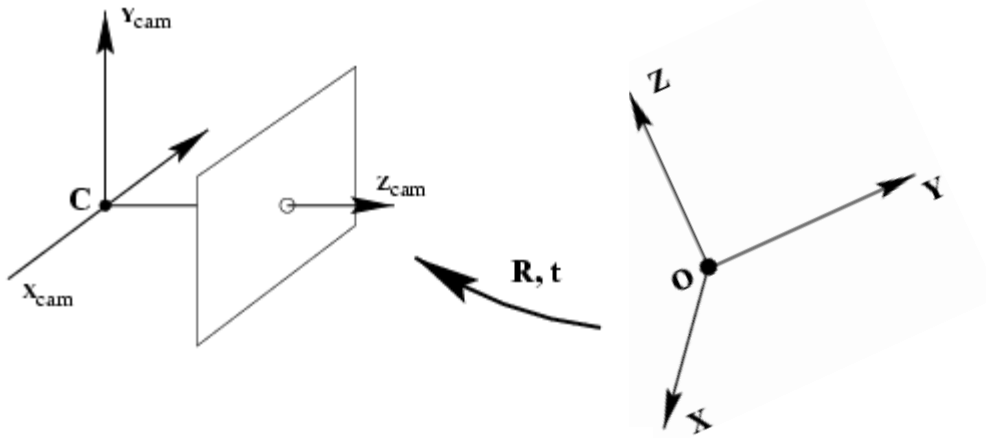
- In non-homogeneous coordinates, the transformation from **world** to normalized **camera** coordinate system is given by:

$$\tilde{X}_{cam} = R(\tilde{X} - \tilde{C}) = R\tilde{X} + t$$

Diagram illustrating the transformation equation:

- $\tilde{X}_{cam}$: coords. of point in normalized camera frame
- R : 3x3 rotation matrix
- $\tilde{X}$: coords. of a point in world frame
- $\tilde{C}$: coords. of camera center in world frame
- t : translation vector

Extrinsic parameters: Rotation and translation



In *non-homogeneous*
coordinates:

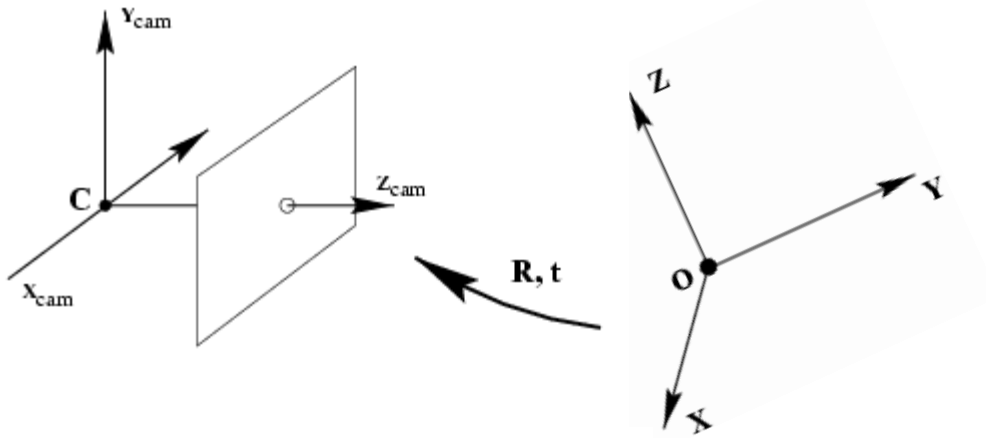
$$\tilde{X}_{\text{cam}} = R\tilde{X} + t$$

In *homogeneous*
coordinates:

$$\begin{pmatrix} \tilde{X}_{\text{cam}} \\ 1 \end{pmatrix} = \begin{bmatrix} R & t \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{pmatrix} \tilde{X} \\ 1 \end{pmatrix}$$

3D transformation
matrix (4 x 4)

Extrinsic parameters: Rotation and translation



In *non-homogeneous*
coordinates:

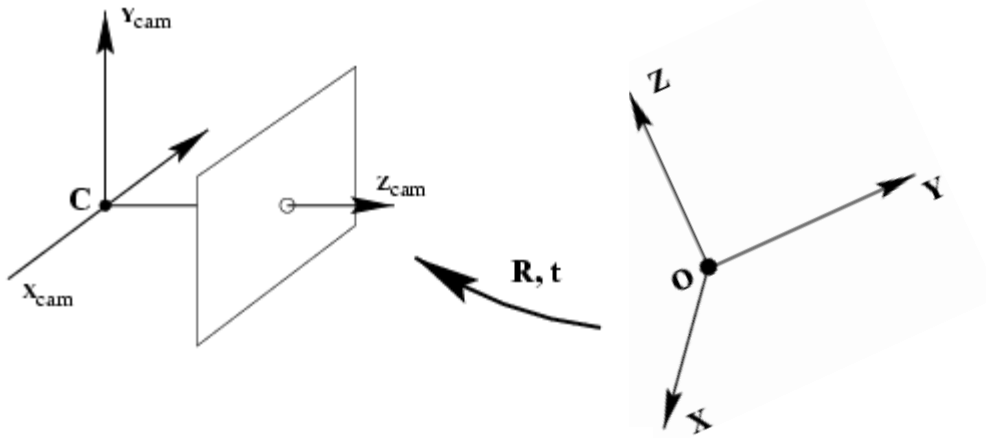
$$\tilde{X}_{cam} = R\tilde{X} + t$$

In *homogeneous*
coordinates:

$$X_{cam} = \begin{bmatrix} R & t \\ \mathbf{0}^T & 1 \end{bmatrix} X$$

3D transformation
matrix (4 x 4)

Extrinsic parameters: Rotation and translation



In *non-homogeneous*
coordinates:

$$\tilde{X}_{\text{cam}} = R\tilde{X} + t$$

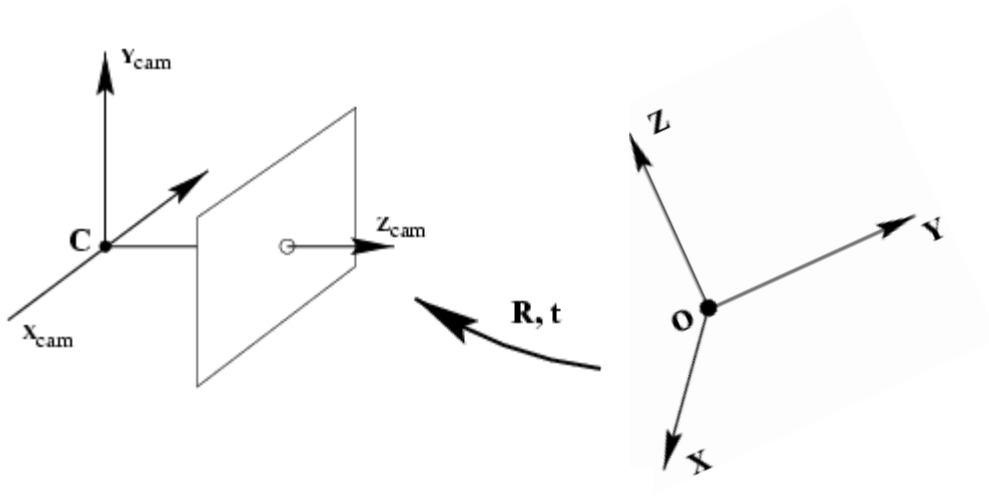
In *homogeneous*
coordinates:

$$\boxed{X_{\text{cam}}} = \begin{bmatrix} R & t \\ \mathbf{0}^T & 1 \end{bmatrix} X$$

Transformation from normalized 3D
coordinates to pixel image coordinates:

$$x \cong K[I|\mathbf{0}]X_{\text{cam}}$$

Extrinsic parameters: Rotation and translation



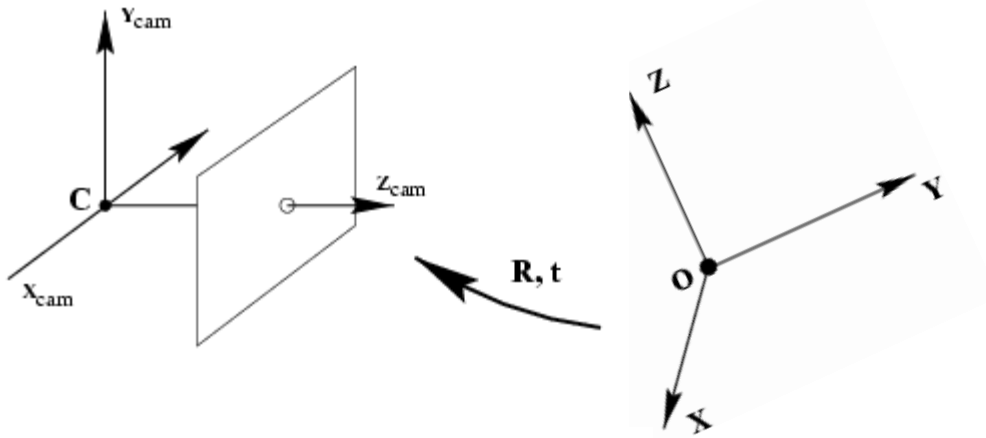
In *homogeneous*
coordinates:

$$x \cong K[I|0] \begin{bmatrix} R & t \\ 0^T & 1 \end{bmatrix} X$$

Finally:

$$x \cong K[R|t]X \quad t = -R\tilde{C}$$

Extrinsic parameters: Rotation and translation



$$x \cong K[R|t]X$$

$$t = -R\tilde{C}$$

coords. of
camera center
in world frame

- What is the projection of the camera center in world coordinates?

$$PC = K[R \mid -R\tilde{C}] \begin{pmatrix} \tilde{C} \\ 1 \end{pmatrix} = K(R\tilde{C} - R\tilde{C}) = 0$$

- The camera center is the **null space** of the projection matrix!

Camera parameters: Summary

$$\mathbf{P} = \mathbf{K}[\mathbf{R}|\mathbf{t}]$$

- Intrinsic parameters
 - Principal point coordinates
 - Focal length
 - Pixel magnification factors
 - *Skew (non-rectangular pixels) – not important in practice*
 - *Radial distortion – important in practice!*

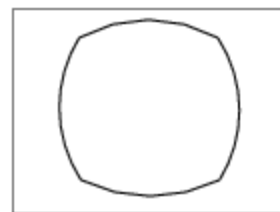
$$\mathbf{K} = \begin{bmatrix} m_x & 0 & 0 \\ 0 & m_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & 0 & \beta_x \\ 0 & \alpha_y & \beta_y \\ 0 & 0 & 1 \end{bmatrix}$$



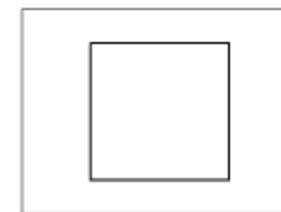
radial distortion



linear image



correction



Summary

- pinhole projection model
- properties of projection
- perspective projection
- orthographic projection
- camera calibration